

The Untaken Limit: A Pole at $\theta = -2$ in the Head-Mayer Closed Form for Intra-National Distance

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Intra-national distance at gravity-consistent $\theta = -5$
on a uniform disk ($R = 1000$ km), vs grid resolution

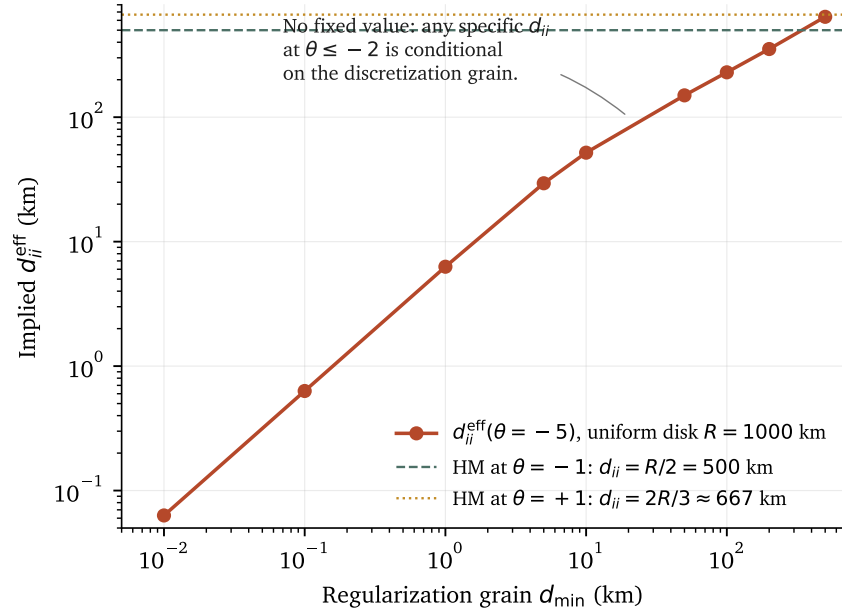


Fig 1. Implied $d_{ii}^{\text{eff}}(\theta = -5)$ on a uniform disk: two-order-of-magnitude span from the regularization grain alone.

The setup. Every gravity regression since 2005 uses CEPII's `distw_harmonic` as the intra-national distance. CEPII constructs it from the Head-Mayer (2002) CES integral $d_{ii}(\theta) = (2/(\theta + 2))^{1/\theta} R$, evaluated at $\theta = -1$. Mayer-Zignago (2011) justify $\theta = -1$ as “the usual coefficient estimated from gravity models”—the reduced-form regression slope, not the structural CES integration parameter $\theta = \delta(1 - \sigma)$, which for $\sigma \in [4, 8]$ and $\delta = 1$ falls in $[-7, -3]$.

The pole. The closed form has a pole at $\theta = -2$: complex-valued for $\theta < -2$. The underlying integral $\int_0^R r^{\theta+1} dr$ diverges in 2D for $\theta \leq -2$. Structurally consistent values fall entirely below the pole.

Closed form at $R = 1000$ km:

$\theta = +1$: 667 km (arithmetic mean)

$\theta = -1$: 500 km (**CEPII**)

$\theta = -1.99$: 9.5 km (near pole)

$\theta = -2$: **POLE**

$\theta = -3, -5, -7$: **complex** (gravity-consistent)

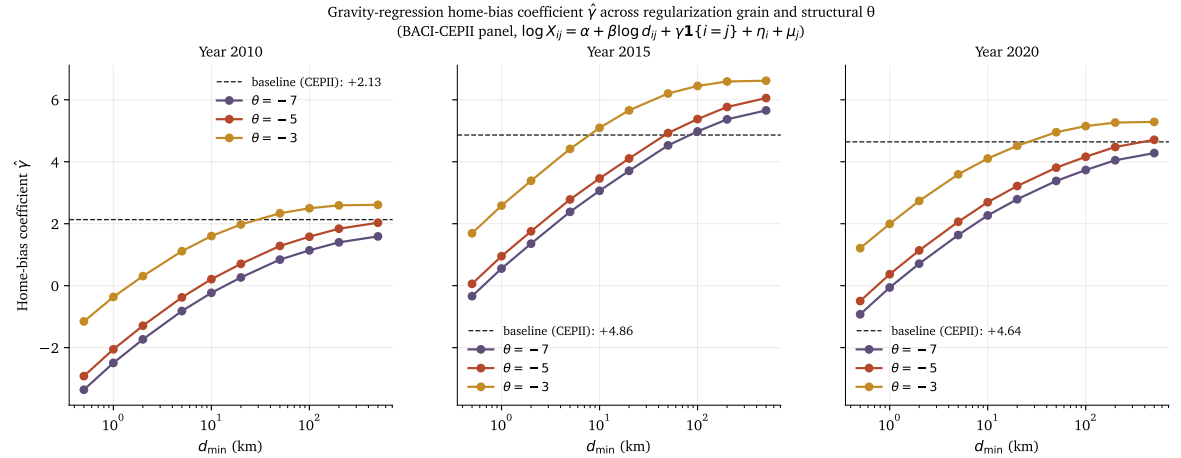


Fig 2. Home-bias coefficient $\hat{\gamma}$ vs the regularization grain, by year and structural θ . CEPII baseline (dashed) at the high- $d_{\min}$ end; sign-flip threshold between 1 and 5 km.

The consequence. At gravity-consistent σ , d_{ii} has no fixed value. The home-bias coefficient inherits the regularization choice $d_{\min}$: from -2.92 (multiplier $0.05\times$) at fine resolution to $+2.13$ (multiplier $8.4\times$, the canonical McCallum finding) at the CEPII baseline. **The border puzzle lives inside the measurement window.**

Takeaway. Border-puzzle multipliers have been reported as structural objects rather than as conditional on a regularization. A $10\times$ multiplier at $d_{\min} = 100$ km and a $5\times$ multiplier at $d_{\min} = 10$ km are not contradictions; they are the same regression conditional on different regularizations. The structurally consistent procedure is to acknowledge that d_{ii} is not a well-defined object at gravity-consistent σ , and to estimate the model in a way that does not pin d_{ii} to a single number.